2. Install green and cool roofs.
3. Use energy-efficient appliances and equipments.
4. Shift all industries away from the urban area.
5. Reduce emission from automobiles.

6.4 Atmospheric Pressure and Wind

Atmospheric pressure is defined as the force per unit area exerted against a surface by the weight of the air molecules above the earth surface. In the Figure below (Figure 6.10), the pressure at point 'X' increases as the weight of the air increases. The atmospheric pressure is not distributed uniformly over the earth. The amount of pressure increases or decreases, according to the amount of molecules, that exerts the force on the surface.

When temperature of the air increases, the air expands and reduces the number of molecules over the unit area. It leads to reduction in pressure. Similarly, when the temperature falls, the air contracts and the pressure increase. Therefore, the temperature and atmospheric pressure are inversely related.

Atmospheric pressure is measured by an instrument called **'Barometer'**.

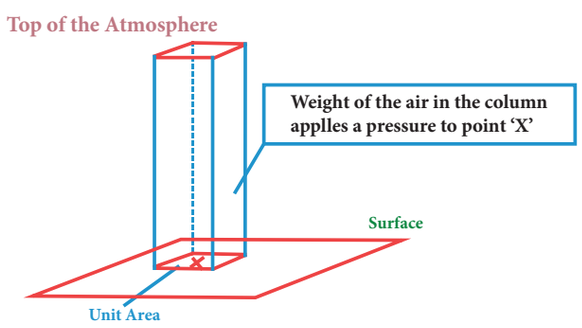


Figure 6.10 Atmospheric Pressure

6.4.1 Vertical Distribution of Atmospheric Pressure

The relationship analysis between altitude and atmospheric pressure is very peculiar. The upper atmosphere is thin and less dense. The pressure at sea level is highest and keeps decreasing rapidly with increasing altitude because of the progressive reduction of the mass above the point where it is measured (Figure 6.11).



Isobar is an imaginary line connecting the places of uniform atmospheric pressure reduced to mean sea level

Brain Storming

People feel discomfort to breathe when they go to the places of higher altitude (mountain sickness). Why?

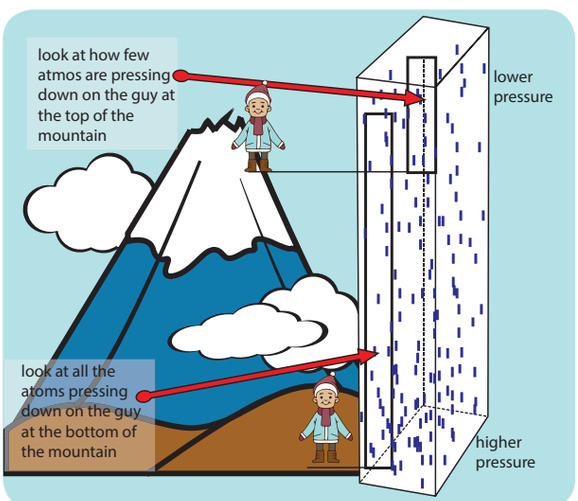


Figure 6.11 Relationship between altitude and pressure

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